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Group-18 & Chemical Bonding

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PART-A

SINGLE ANSWER TYPE

- Which of the following fluorides of xenon on hydrolysis forms XO_3
a) Xe_6 b) XeF_4 c) XeF_2 d) both a&b
- The element which can not form clathrate compounds
a) Xe b) Ar c) Kr d) Ne
- $\text{Xe} + \text{F}_2$ (1: 20) X on reaction with SiO_2 forms
a) $\text{XeOF}_4 \cdot \text{SiF}_4$ b) $\text{XeOF}_2 \cdot \text{SiF}_4$ c) $\text{XeO}_3 \cdot \text{SiF}_4$ d) all the above
- The number of water molecules present in stable xenon gas hydrate
a) 6 b) 4 c) 2 d) 8
- Partial hydrolysis of XeF_6 gives HF and some other compound X. X is
a) XeO_3 b) XeOF_4 c) XeF_2 d) XeF_4
- Platinum on reaction with xenon tetra fluoride forms
a) PtF_6 b) PtF_4 c) $\text{Xe}[\text{PtF}_6]$ d) PtF_2
- Xenon fluorides can act as
a) Oxidising b) Fluorinating c) Fluorine donors d) all the above
- The inert gas lack of triple point
a) Xe b) Ar c) Ne d) He
- Inert gases are soluble in water due to
a) Dipole-dipole interaction. b) Hydrogen bonding
c) Dipole induced dipole interactions d) Vander wall forces
- The I.E. Value of molecular oxygen is equal to
a) Xe b) Ne c) Ar d) He
- The formation of $\text{O}_2^+[\text{PtF}_6]^-$ is the basis for the formation of xenon fluorides. This is because:
A) O_2 and Xe have comparable sizes.
B) both O_2 and Xe are gases.
C) O_2 and Xe have comparable ionisation energies.
D) O_2 and Xe have comparable electronegativities.
- Which of the following gaseous molecules is monoatomic?
A) Chlorine B) Helium C) Oxygen D) Nitrogen
- Which one of the following noble gases is not found in atmosphere?

- A) Rn B) Kr C) Ne D) Ar
14. The inert gas abundantly found in atmosphere is
A) Ar B) Kr C) He D) Xe
15. Of the following species, one which is non-existent:
A) XeF_6 B) XeF_5 C) XeF_4 D) XeF_2
16. Maximum number of compounds are known in the case of
A) Neon B) Xenon C) Krypton D) Argon
17. Which inert gas has abnormal behaviour on liquefaction?
A) Xe B) He C) Ar D) Kr
18. Helium is added to oxygen used by deep sea divers because:
A) It is less soluble in blood than nitrogen under high pressure
B) It is lighter than nitrogen
C) It is readily miscible with oxygen
D) It is less poisonous than nitrogen
19. Hydrolysis of XeF_4 and CaNCN gives respectively:
A) XeO_3 and CaCO_3 B) XeO_2 and CaCN_2
C) XeOF_3 and CaCN_2 D) XeOF_2 and CaCO_3
20. Consider following properties of the noble gases.
I : They readily form compounds which are colourless.
II : They generally do not form ionic compounds.
III: Xenon has variable oxidation states in its compounds.
IV: The smaller He and Ne do not form clathrate compounds.
Select correct properties.
A) I, II, III B) II, III, IV C) I, III, IV D) All
21. Which one of the following configuration represents a noble gas?
A) $1s^2 2s^2 p^6, 3s^2$ B) $1s^2 2s^2 p^6, 3s^1$ C) $1s^2 2s^2 p^6$ D) $1s^2 2s^2 p^6, 3s^2 p^6, 4s^2$
22. In Kroll and I.C.I process of the production of titanium, the inert gas used is
A) Ne B) Ar C) Kr D) Xe
23. Among the following molecules,
(i) XeO_3 (ii) XeOF_4 (iii) XeF_6
those having same number of lone pairs on Xe are:
A) (i) and (ii) only B) (i) and (iii) only C) (ii) and (iii) only D) (i) , (ii) and (iii)
24. XeF_6 on complete hydrolysis gives
A) Xe B) XeO_2 C) XeO_3 D) XeO_4
25. The product of the reaction between one mole of XeO_3 and two mole of XeF_6 is
A) XeO_2F_2 B) XeOF_4 C) XeO_3F_2 D) XeO_4
- MULTIPLE ANSWER TYPE**
26. Select the incorrect order.
A) $\text{He} > \text{Ar} > \text{Kr} > \text{Ne} > \text{Xe}$ – (abundance in air).
B) $\text{He} < \text{Ne} < \text{Ar} < \text{Kr} < \text{Xe}$ – (boiling point).
C) $\text{XeF}_6 > \text{XeF}_4 > \text{XeF}_2$ – (melting point).

- D) $\text{XeF}_6 < \text{XeF}_4 < \text{XeF}_2$ — (Xe – F bond length).
27. Which of the following inert gas(es) form(s) clathrate compound(s) with quinol?
 A) Helium B) Xenon C) Krypton D) Neon
28. Which among the following statements is / are correct?
 A) XeF_4 and SbF_5 combine to form salt.
 B) He and Ne do not form clathrate.
 C) He diffuses through rubber and polyvinyl chloride.
 D) He has lowest boiling point in its group.
29. Thermal decomposition product (s) of XeF_6 is /are
 A) Xe B) XeF_2 C) XeF_4 D) F_2
30. Which of the following statements(s) is /are true for XeF_6 ?
 A) Its partial hydrolysis gives XeOF_4 .
 B) Its reaction with silica gives XeOF_4
 C) It is prepared by the reaction of XeF_4 and O_2F_2
 D) Its reaction with XeO_3 gives XeOF_4 .
31. Which of the following is/are properties of helium?
 A) It is chemically inert. B) It has very high thermal conductivity.
 C) It has extremely low boiling point. D) It has very low viscosity.
32. Select the correct statement(s) regarding the fluorides of xenon.
 A) All three fluorides are decomposed by water, XeF_2 slowly and, XeF_4 and XeF_6 rapidly.
 B) All three fluorides are powerful oxidising agents.
 C) XeF_4 and XeF_6 can act as fluoride ion acceptors as well as fluoride ion donors.
 D) All three fluorides are volatile, readily subliming at room temperature (298 K).
33. $[\text{HXeO}_4]^- + \text{OH}^- \longrightarrow [\text{X}] + [\text{Y}] + \text{O}_2 + \text{H}_2\text{O}$
 The products [X] and [Y] in unbalanced reaction are
 A) Xe B) XeO_3 C) XeO_4 D) $[\text{XeF}_6]^{4-}$
34. Which one of the following reactions of Xenon compounds are feasible?
 A) $3\text{XeF}_4 + 6\text{H}_2\text{O} \longrightarrow 2\text{Xe} + \text{XeO}_3 + 12\text{HF} + 1.5\text{O}_2$
 B) $2\text{XeF}_2 + 2\text{H}_2\text{O} \longrightarrow 2\text{Xe} + 4\text{HF} + \text{O}_2$
 C) $\text{XeF}_6 + \text{RbF} \longrightarrow \text{Rb}[\text{XeF}_7]$
 D) $\text{XeO}_3 + 6\text{HF} \longrightarrow \text{XeF}_6 + 3\text{H}_2\text{O}$
35. XeF_4 and XeF_6 are expected to be
 A) Oxidizing agents B) Reducing agents
 C) Fluorinating agents D) Strongly basic in nature

KEY

1	D	2	D	3	D	4	A	5	B	6	B	7	D	8	D
9	C	10	A	11	C	12	B	13	A	14	A	15	B	16	B
17	B	18	A	19	A	20	B	21	C	22	B	23	D	24	C
25	B	26	AC	27	BC	28	ABCD	29	BCD	30	ABCD	31	ABCD	32	ABCD

33	AD	34	ABC	35	AC										
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PART-B

MULTIPLE ANSWER TYPE

- The pair of molecules with different state of hybridization of the central atom but with the same shape is
 A) $[XeF_2, Br_3^-]$ B) $[CO_2, I_3^-]$ C) $[H_2O, SO_2]$ D) $[XeF_4, [Ni(CN)_4]^{2-}]$
- In which of the following case the hybridization of central atom is sp^3d
 (A) I_3^- (B) SF_4 (C) Solid PCl_5 (D) Solid PBr_5
- Select the correct statement (s) about the compound $NO[BF_4]$:
 a) It has 5 σ and 2 π bond
 b) Nitrogen-oxygen length is higher than nitric oxide (NO)
 c) It is made of diamagnetic species
 d) B – F bond length in this compound is lower than in BF_3
- Which of the following statement is/are correct
 A) O_2 is paramagnetic, O_3 is also paramagnetic
 B) O_2 is paramagnetic, O_3 is diamagnetic
 C) B_2 is paramagnetic, C_2 is also paramagnetic
 D) In $NO \rightarrow NO^+$ and $CO \rightarrow CO^+$, bond lengths decrease
- Which of the following species have identical bond order
 A) CN^- B) O_2^- C) NO^+ D) Cl_2
- Which of the following is correctly arranged in decreasing order of bond angle?
 (A) $NO_2^+ > NO_2 > NO_2^-$ (B) $ClO_2^- > ClO_2^+ > ClO_2$
 (C) $NH_3 > NF_3 > F_2O$ (D) $Cl_2O > H_2O > F_2O$
- Which of the following statements is/are correct for CH_2F_2 ?
 (A) Angle $\angle HCH$ and $\angle FCF$ are not equal (B) It is a polar molecule
 (C) The central carbon atom is sp^3 hybridised
 (D) $\angle FCF$ is more than standard tetrahedral bond angle while $\angle HCH$ is less than standard tetrahedral bond angle.

8. Which of the following molecular orbitals has two nodal planes?
 (A) Π_{2Py} (B) Π_{2Px}^* (C) σ_{2Pz}^* (D) Π_{2Py}^*
9. In which of the following all the atoms are in the same plane ?
 A) NF_3 B) $N(SiH_3)_3$ C) Benzene, C_6H_6 D) $CHF = C = CHF$
10. Which of the following is/are correct about the bond angle?
 (A) $OF_2 < H_2O < Cl_2O < ClO_2$ (B) $PF_3 < PCl_3 < PBr_3 < PI_3$
 (C) $NH_3 > NF_3$ (D) $KrF_4 < SF_2 < N_2H_2$
11. Which of the following is/are correct?
 (A) carbon-carbon bond length in CaC_2 will be more than in $CH_2 = C = CH_2$
 (B) O – O bond length in KO_2 will be more than in Na_2O_2
 (C) O – O bond length in $O_2[PtF_6]$ will be less than that in KO_2
 (D) N – O bond length in NO gaseous molecule will be smaller than in NOCl gaseous molecule

INTEGER TYPE

12. Number of hydrogen bonds in $H_9O_4^+$ species is
13. How many orders are correct match?
 (i) Boiling point : $PH_3 < AsH_3 < NH_3 < SbH_3$
 (ii) Dipole moment: $CH_3F > CH_3Cl > CH_3Br$
 (iii) Bond angle: $H_2O < OF_2 < OCl_2$
 (iv) Solubility: $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$
 (v) Acidic strength: $HCOOH > C_6H_5COOH > CH_3COOH > C_6H_5OH > CH_3OH > H_2O$
14. The ratio of lone pairs between $AlCl_4^-$ and CO_2 is
15. Out of $I_3^-, ICl_2^-, BeCl_2, XeF_2, XeF_6, BrF_5, HC \equiv CH$. How many will have linear shape?
16. The number of σ and π bonds are present in Hexa cyano ethane is $5x$. Then x is
17. In I_5^- ion number of lone pairs are present on central atom is
18. Find the number of molecules or ions in which d-orbitals is/are not used in hybridization:
 $PCl_6^-, PCl_4^+, IF_4^-, IF_5, XeO_3F_2, ICl_2^+, SF_2, SF_6, AsF_4^+, SiF_4$

- $$(\text{CH}_3)_2\text{NH}_2, \text{CH}_3\text{OH}, \text{C}_6\text{H}_6, \text{C}_6\text{H}_5\text{NH}_2, \text{CH}_3\text{NH}_2, \text{Na}_2\text{SO}_4, \text{BaSO}_4$$

KEY

[illegible]